Maekawa's theorem and the linear constraints of flat folding

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Abstract

A flat-folded state of a crease pattern is a layer order between overlapping facets. Of the conditions that make a layer order valid, the taco–tortilla and taco–taco constraints are linear over $\mathbb{F}_2$, while transitivity is not. We determine what the linear constraints at a vertex say about its mountain–valley assignment. At every vertex, Maekawa's rule modulo four is the $\mathbb{F}_2$-sum of all the linear constraints there, and the assignments induced by their solutions form the affine hull over $\mathbb{F}_2$ of the flat-foldable ones: Hull's crimping recursion with every condition read modulo four. At a generic vertex this hull is the set of flat-foldable assignments itself. There the taco–tortilla constraints pair up the creases, forcing equal assignments on the two creases at the extreme rays of the folded image and opposite ones on every other pair, so that every solution, transitive or not, satisfies Maekawa's rule $\boldsymbol{M - V = \pm 2}$ on the numbers of mountain and valley creases. The pairs are the persistence pairs of the folded angle, read as a height function around the vertex, and Hull's recursion cancels them in order of increasing persistence. The proofs read the layer order from an extreme crease, where every constraint becomes homogeneous, and rest on one structural fact: every overlapping pair of non-adjacent sectors occurs in exactly two constraints, and every adjacent pair in one. They do not use Maekawa's theorem, and prove it anew, with transitivity entering only to order the folds that coincide in the folded image.

Keywords: Flat folding, Maekawa's theorem, Layer order, Linear constraints over $\mathbb{F}_2$, Crimping, Persistent homology

MSC Classification: 52C99 , 05C65 , 55N31 , 68U05

1 Introduction

A flat-folded state of a crease pattern is not determined by its creases: one must also say, wherever two facets land on each other, which of them lies above. Justin gave the conditions under which such layer orders describe a folding that does not pass through itself [1, 2], and Akitaya, Demaine and Ku stated them facewise, for patterns with convex facets, as a finite list of constraints on the order between pairs of overlapping facets [3]. Two of these families, taco–tortilla and taco–taco, are linear when the layer order is written over $\mathbb{F}_2$ (Fig. 1). Ku, Terao and Terao treat them as a linear system over $\mathbb{F}_2$ [4]; Nakajima, who wrote the same constraints as an Ising model [5], uses their cycles of odd parity to certify that a pattern cannot fold flat [6], and has enumerated the flat-folded states of single vertices with random angles [7]. The third family, transitivity, is not linear; the linear families alone are solved by Gaussian elimination, and transitivity is where the NP-hardness of flat foldability lies [8, 9].

The classical local rules of flat folding live on the mountain–valley assignment instead. Maekawa's rule says that the numbers of mountain and valley creases at an interior vertex differ by two [10–12], the big–little–big lemma that the two creases bounding a sector smaller than both its neighbours have opposite assignments [2, 12], and Hull's crimping recursion decides exactly which assignments of a single vertex fold flat [13]. Since the assignment of a crease is determined by the order of the two facets it separates [3], all these are statements about layer orders. This paper asks how much of them the linear constraints at a vertex already contain, and answers the question completely.

Our first result, Theorem 1, is an identity. Written in the layer variables, Maekawa's rule modulo four is a parity condition around each interior vertex, the *vertex row*, and Theorem 1 says that the vertex row is exactly the sum over $\mathbb{F}_2$ of the taco–tortilla and taco–taco constraints at that vertex. It rests on a structural fact, Lemma 2: every overlapping pair of non-adjacent sectors occurs in exactly two constraints at the vertex, and every adjacent pair in one. In the sum of all the constraints only the adjacent pairs survive, and the constant term depends on how the variables are written. Numbering the sectors from a crease at the lowest ray of the folded image, which no sector covers, makes every constraint homogeneous, and the constant comes from the one adjacent pair that this numbering writes backwards. Appendix A gives three other proofs. One of them draws the folded sectors as half-circles on one side of a line, which turns the folded vertex into a closed plane curve that crosses itself, and applies Whitney's formula [14]: the rotation number of a closed curve has the parity of its double points plus one. The classical proof of Maekawa's rule also counts how a curve turns around the vertex, but uses the cross-section of the folded paper, which does not cross itself [11, 15]; Figs. 2(b) and 6(b) draw the two curves of one vertex.

Our second result, Theorem 2, goes beyond parity at *generic* vertices, those at which no two crease rays coincide in the folded image, and more generally at vertices without taco–taco constraints. There the taco–tortilla constraints pair up the creases: the two creases of a pair are joined by a chain of constraints that forces them to have equal assignments, or opposite ones, in every solution. The two creases at the extreme rays of the folded image form a pair forced equal, and every other pair is forced opposite. Hence every assignment of the layer variables that satisfies the taco–tortilla

constraints at a generic vertex, transitive or not, has $M - V = \pm 2$, the sign being the common assignment of the extreme creases, and the shortest chain is the big–little–big lemma. The proof reads the layer order as for Theorem 1: the extreme creases cut the sectors into two sides, and only the chain that joins them has a sector on each side. The chains make precise, at a single vertex, how non-penetration determines local stacking orders one after another [7].

The pairs have a meaning that does not mention the constraints (Theorem 3). Read the angle of the folded image as a height function on the circle of paper around the vertex. The creases are its critical points, and its persistent homology [16, 17] pairs every minimum with the maximum at which the arc that the minimum starts, below a rising level, merges with an older one. These persistence pairs are exactly the pairs of Theorem 2, the extreme creases forming the essential pair. Each pair can be pictured as a lake, filled from its minimum until it overflows at its maximum (Fig. 3c), and Hull's crimping recursion, which crimps a smallest sector at every step, cancels the pairs in order of increasing persistence.

Our last result, Theorem 4, describes the linear constraints at an arbitrary vertex. The assignments they allow form the affine hull over $\mathbb{F}_2$ of the flat-foldable ones: Hull's crimping recursion with every condition taken modulo four. At a generic vertex this hull contains nothing else, but whenever the recursion crimps a run of two or more equal sectors, or ends at a vertex of degree six or more with all angles equal, it does: at the degree-four vertex with angles $45°, 135°, 135°, 45°$ the linear constraints allow the assignment VMVV, which satisfies Maekawa's rule but does not fold flat, and at the regular vertex of degree six they allow every crease to be a mountain (Examples 3 and 4). At a single vertex, this is where transitivity is needed, and the only place. The proof rests again on Lemma 2. The relations that the constraints impose on the assignment are the components of a hypergraph (Lemma 3), and crimping a smallest run of sectors never merges two components (Lemma 4); a dimension count against Hull's recursion closes the argument.

Theorems 1 and 2 are special cases of Theorem 4, but their proofs are independent of it, and of Maekawa's theorem, on which Hull's recursion and therefore Theorem 4 rest. Every flat-folded state of a vertex is a solution of the taco–tortilla constraints of a nearby generic vertex (Proposition 3), transitivity serving only to order the folds that coincide in the folded image, and Theorem 2 then gives a proof of Maekawa's theorem from the linear constraints (Corollary 3).

Section 2 sets up the layer variables and the vertex row, and Sec. 3 the folded vertex and its constraints. Section 4 proves Lemma 2 and Theorem 1, Sec. 5 introduces the hypergraph of the constraints, Sec. 6 proves Theorem 2 and states Theorem 3, and Sec. 7 proves Theorem 4. Section 8 derives Maekawa's theorem and collects consequences for whole crease patterns: the linear system of a pattern already satisfies the conditions of Theorem 4 at every vertex, and adding Maekawa's rule modulo four never changes its rank or its solutions. Appendix A gives three other proofs of Theorem 1, Appendix B a proof of Theorem 2 by crimping and the proofs of Proposition 1 and Theorem 3, and Appendix C the computations with which we checked every statement.

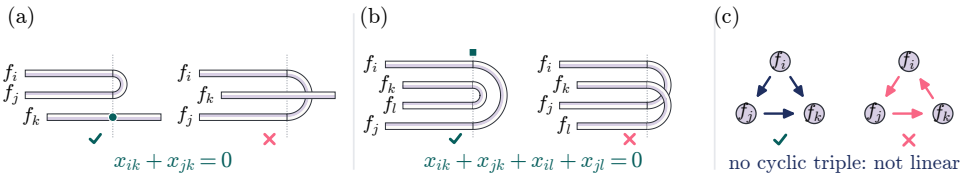


Fig. 1 The conditions on a layer order, the first two seen edge-on. Each layer shows its white side on one face and its lilac side on the other, and the dotted line marks the crease. (a) Taco–tortilla: the facets f_i and f_j are folded along their common crease (the taco), and the facet f_k (the tortilla) covers the crease line in its interior. It may lie below both (left) or above both, but not between them (right). The dot, where the crease line meets the tortilla, is the symbol of a taco–tortilla row in the other figures. (b) Taco–taco: two tacos folded along the same line may be nested (left) or stacked, but not interleaved (right); the square is the symbol of a taco–taco row. (c) Transitivity: three facets with a common region are stacked (left), never ordered cyclically (right). The first two conditions are linear over $\mathbb{F}_2$, the third is not

2 Layer orders and their linear constraints

Let G be a crease pattern on a sheet of paper, a closed disk, with convex facets $f_1, f_2, \ldots$, every interior vertex of even degree and satisfying Kawasaki's condition, so that the folding map μ, obtained by reflecting across the creases, is well defined [12]. Two facets overlap if their images meet in positive area. The layer variable of an overlapping pair is $r_{ij} = +1$ if f_i lies above f_j and $r_{ij} = -1$ otherwise, with $r_{ji} = -r_{ij}$; over $\mathbb{F}_2$ we write $r_{ij} = (-1)^{x_{ij}}$, so that $x_{ji} = x_{ij} + 1$. Among the facewise conditions of Ref. [3], two are linear (Fig. 1):

(TT) *taco–tortilla*: if e_{ij} is the crease between f_i and f_j and the image of a facet f_k covers a segment of $\mu(e_{ij})$ in its interior, then $r_{ik} = r_{jk}$, that is $x_{ik} + x_{jk} = 0$;

(TC) *taco–taco*: if the images of two creases e_{ij}, e_{kl} overlap in a segment and the four facets have a common region of positive area, then $r_{ik}r_{jk}r_{il}r_{jl} = 1$, that is $x_{ik} + x_{jk} + x_{il} + x_{jl} = 0$.

Every flat-folded state satisfies both. We call these equations the *rows* of the crease, or of the pair of creases; a row is *at* a vertex v when all its creases and facets are incident to v. Sums of rows are taken after writing every variable in one fixed orientation of its pair, using $x_{ji} = x_{ij} + 1$. The third condition of Ref. [3], transitivity, asks that no three facets with a common region be ordered cyclically, and it is not linear. For patterns with convex facets the three conditions characterise the flat-folded states [3].

Mountain and valley.

The facets can be coloured $p_i = \pm 1$, the side of the paper that f_i shows once folded, with $p_i = -p_j$ across every crease. Seen from the side on which f_i is face up, a mountain fold sends its neighbour below and a valley fold above, so the assignment $\sigma_e = \pm 1$ (mountain, valley) of the crease e_{ij} in a flat-folded state is

$$\sigma_{e_{ij}} = p_i \, r_{ij}, \tag{1}$$

the same for both orderings of the pair since $p_j r_{ji} = p_i r_{ij}$. This is the facewise form of the observation, used in Ref. [3], that the assignment of a crease prescribes the order

of its two facets. We use Eq. (1) to define σ for every assignment of the layer variables, transitive or not.

The vertex row.

Let v be an interior vertex of degree $n = 2m \geq 4$ and $f_1, \ldots, f_n$ the facets around it in cyclic order, the *sectors* of v, with f_k and f_{k+1} separated by the crease e_k (indices modulo n). Identifying $\sigma = \pm 1$ with a bit, the assignments of the creases at v form the space $\mathbb{F}_2^n$. The *holonomy* of v is $W_v = \prod_k r_{k,k+1}$, and the *vertex row* is the equation

$$W_v = -1, \qquad \text{that is} \qquad \sum_{k=1}^{n} x_{k,k+1} = 1. \tag{2}$$

Multiplying Eq. (1) around v, and using that the p_k alternate so that $\prod_k p_k = (-1)^m$, gives $\prod_k \sigma_{e_k} = (-1)^m W_v$, while $\prod_k \sigma_{e_k} = (-1)^V$ if V creases are valleys. Hence

$$W_v = -1 \iff V \not\equiv m \pmod 2 \iff M - V \equiv 2 \pmod 4, \tag{3}$$

with $M = n - V$ the number of mountains. The vertex row is Maekawa's rule modulo four. At degree four, where $|M - V| \leq 4$, it is Maekawa's rule $M - V = \pm 2$; at higher degree it is weaker.

3 The folded vertex

Near $\mu(v)$ the image of the sector f_k is a wedge, which we describe by an angular interval I_k. Measuring angles in the folded image from the image of e_n,

$$c_0 = 0, \qquad c_k = c_{k-1} + (-1)^{k+1}\alpha_k, \tag{4}$$

where $\alpha_k > 0$ is the angle of f_k; then I_k is the interval between c_{k-1} and c_k, and the crease e_k lies along the ray c_k. Kawasaki's condition is exactly $c_n = c_0$. Every c_k is the sum of the odd angles up to k minus the sum of the even ones; both sums grow with k from 0 to π, so any two of the c_k differ by less than π, a difference of π requiring a single sector of angle π. The folded image of the vertex therefore lies in a wedge of angle less than π, its angles can be read on the real line, and two sectors overlap exactly when their intervals meet in more than a point. The walk turns at every crease: odd creases are local maxima, where I_k and I_{k+1} both extend towards smaller angles, and even creases local minima. The vertex is *generic* if $c_1, \ldots, c_n$ are pairwise distinct. Adjacent sectors of a generic vertex then have different angles, since two adjacent intervals share an endpoint.

Lemma 1 (Local rows) *The taco–tortilla rows at v are the pairs (e_k, f_g) with c_k in the interior of I_g. The taco–taco rows at v are the pairs of creases (e_k, e_l) with $c_k = c_l$ that are both maxima or both minima. In particular a generic vertex has no taco–taco row.*

Proof The image of a convex facet with a corner at v is a convex polygon with a corner at $\mu(v)$, so it lies in the cone of its angular interval and contains a neighbourhood of $\mu(v)$ in that cone. It therefore covers a segment of the ray c_k in its interior exactly when c_k is interior to its interval. Two crease images through $\mu(v)$ overlap in a segment exactly when they lie on the same ray; the four facets then have a common region if and only if they all lie on the same side of the ray, that is if both creases are maxima or both minima of the walk. □

Flat-foldable assignments.

The creases at v, restricted to a small disk around v, form a single-vertex crease pattern whose facets are the sectors. It folds flat, by Kawasaki's condition [12], and its flat-folded states are the orders of the sectors that satisfy the rows at v and transitivity [3]; we call them the flat-folded states of v. A mountain–valley assignment of the creases at v is *flat-foldable* if a flat-folded state of v induces it, and we write $W_{\text{flat}}(v) \subseteq \mathbb{F}_2^n$ for the set of these assignments. Every flat-folded state of G restricts to a flat-folded state of each of its interior vertices.

By Lemma 1, the rows at v depend on v only through its walk. We use them, with the rows given by Lemma 1, for every closed walk (4) with positive steps, whether or not the angles add up to 2π; such walks describe the vertices of cones [13]. Lemma 2 and Theorem 1 below hold for them with the same proofs, Lemma 2 for walks of length at least four; at length two the two sectors are adjacent at both creases, there is no row, and the vertex row $x_{12} + x_{21} = 1$ holds identically. Secs. 6 and 7 use closed walks in this generality.

Example 1 The vertex of Fig. 2 has degree six and sector angles $40^\circ, 70^\circ, 100^\circ, 60^\circ, 40^\circ, 50^\circ$. Its walk is $(c_1, \ldots, c_6) = (40, -30, 70, 10, 50, 0)$ degrees, which is generic, with $I_1 = [0, 40]$, $I_2 = [-30, 40]$, $I_3 = [-30, 70]$, $I_4 = [10, 70]$, $I_5 = [10, 50]$ and $I_6 = [0, 50]$; every two sectors overlap. Its twelve taco–tortilla rows are (e_1, f_g) for $g = 3, 4, 5, 6$; (e_4, f_g) for $g = 1, 2, 3, 6$; (e_5, f_3), (e_5, f_4), (e_6, f_2) and (e_6, f_3). The creases e_2 and e_3, at the lowest and highest rays, lie inside no sector. On the fifteen layer variables these rows have rank 11, hence 16 solutions; 12 of them are transitive, and these are the flat-folded states of the vertex [3]. One of them is drawn in Fig. 2(b).

4 The sum of the rows

The pairs $\{k, k+1\}$ of sectors, one for every crease, are *adjacent*. Every row consists of overlapping pairs of sectors, and the rows at a vertex contain each of them a fixed number of times.

Lemma 2 (Two rows per pair) *Every adjacent pair of sectors lies in exactly one row at v, and every other overlapping pair in exactly two.*

Proof Let $\{a, b\}$ be an overlapping pair, with $I_a \cap I_b = [\ell, u]$ and $\ell < u$. A taco–tortilla row contains $\{a, b\}$ when a crease of one of the two sectors lies in the interior of the interval of

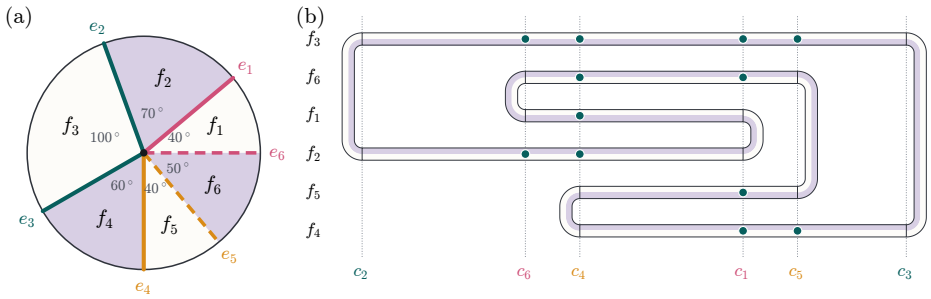


Fig. 2 The vertex of Example 1. (a) The crease pattern, each sector tinted by the side of the paper it shows once folded, with the mountain (solid) and valley (dashed) creases of the flat-folded state in (b). (b) That state seen edge-on, over the angular coordinate of the folded image: the sector f_k is a layer over its interval I_k, and the crease e_k is a fold at the angle c_k (dotted lines). The layer order is $f_3, f_6, f_1, f_2, f_5, f_4$ from top to bottom, and the assignment of $e_1, \dots, e_6$ is MMMMVV, with $M - V = 2$. A dot where a dotted line meets a layer in its interior is a taco–tortilla row (Lemma 1); there are twelve, and they do not depend on the state. Followed around the vertex, the layers and folds form the cross-section of the folded paper, a simple closed curve. The creases are coloured by the pairs of Theorem 2 (Example 2): the extreme creases e_2, e_3 in teal, e_4, e_5 in amber, e_6, e_1 in rose

the other, and a taco–taco row when a crease of each lies on one ray, both maxima or both minima, so that the two intervals extend from it on the same side. In either case the crease lies at ℓ or at u (Fig. 3a). If only one of the two intervals has its lower endpoint at ℓ, the crease at that endpoint lies in the interior of the other interval, and gives one taco–tortilla row; if both do, at distinct creases, these are two minima on one ray, and give one taco–taco row; if both do at the same crease, a and b are its two sectors, and no row at ℓ contains $\{a, b\}$. The same holds at u. Two sectors share at most one crease, and adjacent ones exactly one, since $n \geq 4$. □

Theorem 1 *Let v be an interior vertex. As affine equations over $\mathbb{F}_2$, the vertex row (2) is the sum of all the taco–tortilla and taco–taco rows at v.*

Proof Let e_B be a crease at the lowest ray of the folded image, and number the sectors cyclically from it, so that $B = n$: the sector f_1 follows e_n and f_n precedes it. Renumbering changes neither the rows nor the vertex row. Write every layer variable as x_{ij} with $i < j$, using $x_{ji} = x_{ij} + 1$.

Every row is then homogeneous. No sector contains the lowest ray in the interior of its interval, so no taco–tortilla row is at e_n, and a taco–tortilla row (e_k, f_g) has $k < n$: its taco sectors f_k and f_{k+1} are consecutive, and g, being neither of them, is smaller than both or larger than both. The row is $x_{kg} + x_{k+1,g} = 0$ or $x_{gk} + x_{g,k+1} = 0$, that is, the row of Sec. 2 with no variable or both variables reversed, and its constant is 0. A taco–taco row (e_k, e_l) with $k < l$ has four distinct sectors. If $l < n$, then $k + 1 < l$, and its four variables $x_{kl}, x_{k+1,l}, x_{k,l+1}, x_{k+1,l+1}$ are all written with $i < j$. If $l = n$, then $1 < k$, and two of them, x_{k1} and $x_{k+1,1}$, are reversed, which adds $1 + 1 = 0$.

Summing all the rows, by Lemma 2 the variable of every non-adjacent pair occurs twice and cancels, and every adjacent variable occurs once. The sum is therefore the homogeneous

equation

$$\sum_{k=1}^{n-1} x_{k,k+1} + x_{1n} = 0,$$

in which only the adjacent pair of e_n is written backwards. Since $x_{1n} = x_{n1} + 1$, this is $\sum_k x_{k,k+1} = 1$, the vertex row. □

The proof uses the position of e_n only for the taco–tortilla rows; for the taco–taco rows the numbering may start anywhere. At the vertex of Example 1 the lowest crease is e_2, and the numbering runs $f_3, f_4, f_5, f_6, f_1, f_2$ (Fig. 3b); the regular vertex of Example 4 below is already numbered from a lowest crease. Appendix A gives three other proofs of Theorem 1. They write the variables in a fixed orientation, compute the constant of the sum of the taco–tortilla rows at a generic vertex as the parity of a count of crossings and nestings of intervals, and fix it by Whitney's formula for a closed curve built from the folded vertex, by moving the crease positions, or on a single flat-folded state.

5 The hypergraph of the rows

The rows at a vertex v form a hypergraph H_v. Its vertices are the overlapping pairs of sectors; a taco–tortilla row (e_k, f_g) is a hyperedge with the two vertices $\{k, g\}, \{k+1, g\}$, and a taco–taco row (e_k, e_l) one with the four vertices $\{k, l\}, \{k+1, l\}, \{k, l+1\}, \{k+1, l+1\}$. Reading constraints on two variables as the edges of a graph on pairs of facets is the graphical representation of Ref. [6]; the taco–taco rows add hyperedges with four vertices. By Lemma 2, the adjacent pairs lie in one hyperedge each, and every other vertex in two.

Let $W_{\text{lin}}(v) \subseteq \mathbb{F}_2^n$ be the set of mountain–valley assignments induced, through Eq. (1), by the solutions of the rows at v. Since flat-folded states of v satisfy the rows, $W_{\text{flat}}(v) \subseteq W_{\text{lin}}(v)$. Call a *component* of H_v a connected component of the hypergraph, and let $\gamma(v)$ be the number of components that contain adjacent pairs.

Lemma 3 (Relations are components) *$W_{\text{lin}}(v)$ is an affine subspace of $\mathbb{F}_2^n$ of dimension $n - \gamma(v)$. It is cut out by one equation for every component C containing adjacent pairs, the sum of the rows of C, which reads $\sum_{\{k,k+1\}\in C} x_{k,k+1} = \kappa_C$ with $\kappa_C \in \mathbb{F}_2$.*

Proof By Eq. (1), the assignment of e_k is the variable $x_{k,k+1}$ up to the constant p_k, so $W_{\text{lin}}(v)$ is the image of the solution space of the rows under the projection onto the adjacent variables. The rows have solutions, the flat-folded states of v, and the image of an affine solution space under a coordinate projection is cut out by those sums of rows in which only the retained coordinates survive. A sum of a set S of rows has no non-adjacent variable exactly when every non-adjacent pair lies in an even number of rows of S, which by Lemma 2 means in both or in neither of its two rows: S is a union of components. The sum of the rows of a component C contains every adjacent variable of C once, again by Lemma 2, and different components contain different adjacent pairs. The equations of the components containing adjacent pairs are therefore independent, and they cut out $W_{\text{lin}}(v)$. □

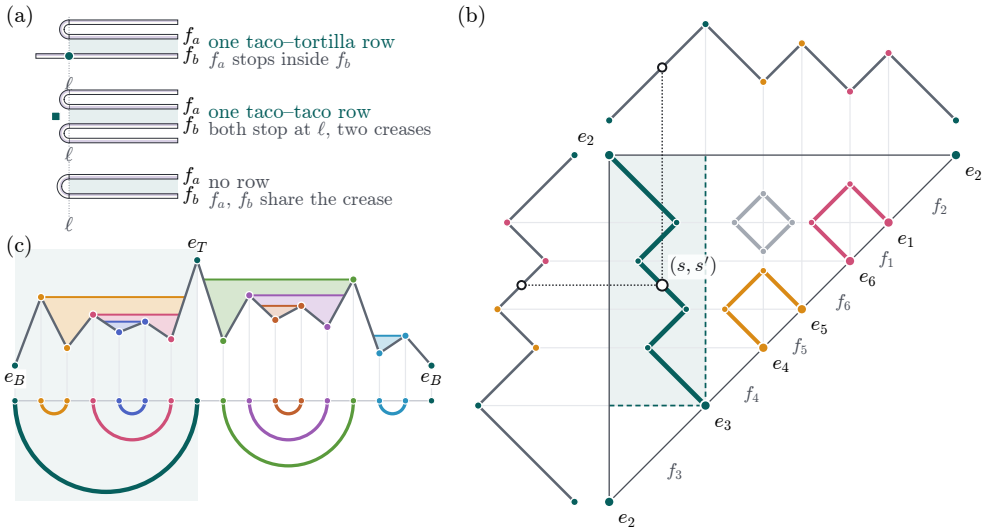


Fig. 3 The rows of a vertex. (a) The rows that contain a pair of sectors f_a, f_b lie at the two ends of the intersection of their intervals (shaded), one at each end (Lemma 2). At the lower end ℓ either one interval stops inside the other, and the crease there is in a taco–tortilla row with the other sector as tortilla (dot); or both stop at two creases on the ray of ℓ, which have a taco–taco row (square); or the two sectors share the crease at ℓ, and no row contains them there. (b) The vertex of Example 1, read from its lowest crease e_2. Top: the angle of the folded image along the paper, with the sectors in the order $f_3, f_4, f_5, f_6, f_1, f_2$ of the proof of Theorem 1; left: the same along the other axis. A point (s, s') of the triangle $s < s'$ is a pair of points of the paper, and it lies on the coloured curve when the two points have the same image (dotted). Every overlapping pair of sectors is a segment of the curve, every taco–tortilla row a point where it crosses a crease line (dots), and every adjacent pair an end on the diagonal, at a crease: the curve is the row graph R_v. The extreme crease e_3 cuts off the rectangle (shaded) of the pairs with one sector on each side, which contains only the path from e_3 to e_2; the corner of the rectangle is again e_2. The two other paths pair e_4 with e_5 and e_6 with e_1, and the closed curve (grey) is the cycle of Example 2; colours as in Fig. 2. (c) A generic vertex of degree sixteen, with sector angles 39, 29, 19, 10, 6, 10, 45, 46, 26, 14, 8, 12, 27, 42, 10, 17 degrees: the heights of its creases, in their order around the vertex from e_B, and its eight pairs as arcs. Every pair joins a maximum to a minimum, the pairs are nested (Proposition 1), and the pair of the extreme creases encloses those of the first side (shaded). Above, every other pair is a lake (Theorem 3): the basin of its minimum, filled up to the level of its maximum, where it would overflow into the basin of a lower minimum; the lakes of nested pairs lie inside one another

The sum of all the components is the sum of all the rows, which by Theorem 1 is the vertex row. At a vertex without taco–taco rows, in particular at a generic vertex, every row has two vertices, and H_v is a graph, the *row graph* R_v (Fig. 3b). By Lemma 2 its components are cycles and paths whose ends are adjacent pairs; there are n adjacent pairs, hence m paths, and $\gamma(v) = m$. A path from $\{a, a+1\}$ to $\{b, b+1\}$ *pairs* the creases e_a and e_b, and its equation $x_{a,a+1} + x_{b,b+1} = \kappa_P$ says, by Eq. (1), that every solution has

$$\sigma_{e_a}\,\sigma_{e_b} = \varepsilon_P, \qquad \varepsilon_P = p_a p_b\,(-1)^{\kappa_P}; \tag{5}$$

we say that the path forces equal or opposite assignments on its two creases as $\varepsilon_P = +1$ or -1.

6 The pairing at a generic vertex

A crease at the highest ray of the folded image is a maximum, and a crease at the lowest ray a minimum. At a vertex without taco–taco rows, in particular at a generic vertex, each of the two rays carries a single crease, since two maxima on one ray, or two minima, would have a taco–taco row. We call these two creases e_T and e_B the *extreme creases*; no sector contains them in the interior of its interval, and no row is at them.

Theorem 2 *At an interior vertex of degree $2m$ without taco–taco rows, in particular at a generic vertex, the taco–tortilla rows pair the creases into m pairs, each made of a maximum and a minimum. The two extreme creases form a pair, forced to have equal assignments, and the other $m-1$ pairs are forced to have opposite ones. In particular, every assignment of the layer variables between the sectors at v that satisfies the taco–tortilla rows at v has $M-V=\pm 2$, with $M-V=2$ if the extreme creases are mountains and $M-V=-2$ if they are valleys.*

The second statement follows from the first: the pairs partition the creases, every opposite pair contributes 0 to $M-V$, and the equal pair ± 2. The assignment need not be transitive.

Proof The paths of R_v pair the creases (Sec. 5). We show where they run, which creases they pair, and which relations they force.

Sides. The extreme creases cut the cyclic sequence of sectors into two *sides*, $(f_{B+1},\dots,f_T)$ and $(f_{T+1},\dots,f_B)$. A row (e_k,f_g) has $k\neq T,B$, so its taco sectors f_k and f_{k+1} lie on one side, and its two vertices $\{k,g\}$ and $\{k+1,g\}$ have the same number of sectors on the first side: none, one or two. Every path of R_v keeps this number. Only two adjacent pairs, those of e_T and e_B, have one sector on each side, so the path that starts at the adjacent pair of e_T ends at that of e_B, and every other path joins two creases of one side.

Maxima and minima. The rows containing a pair of sectors lie at the two ends of the intersection of their intervals, one at each end (proof of Lemma 2), and a row (e_k,f_g) lies at the upper end of the intersection for both its vertices if e_k is a maximum, at the lower end for both if e_k is a minimum. A path enters each of its inner vertices at one end and leaves it at the other, so its rows alternate between maxima and minima. An adjacent pair $\{a,a+1\}$ has its only row at the end opposite to c_a: at a minimum if e_a is a maximum, and conversely. Call a pair of sectors *mixed* if one of them ends at a maximum and the other at a minimum. Adjacent pairs are mixed, and every row (e_k,f_g) has one mixed vertex and one unmixed, since f_k and f_{k+1} end at creases of different types; a path, which starts and ends at a mixed pair, therefore has an even number of rows. If it starts at a maximum, its first row is at a minimum and its last row at a maximum, through which it enters its last vertex at the upper end; the crease of that adjacent pair, at the lower end, is a minimum. Every pair is made of a maximum and a minimum.

Relations. Number the sectors from e_B and write the variables as in the proof of Theorem 1, so that every row reads $x_u=x_w$: all the variables of a path are equal. A path joining e_a and e_b on one side has $a,b\neq n$ and gives $x_{a,a+1}=x_{b,b+1}$, both adjacent variables written forwards; since $a-b$ is odd, $p_ap_b=-1$, and Eq. (1) gives $\sigma_{e_a}\sigma_{e_b}=-1$. The path of the extreme creases gives $x_{T,T+1}=x_{1n}=x_{n1}+1$, that is $r_{T,T+1}=-r_{n,n+1}$, and since $T-n$ is odd, $\sigma_{e_T}\sigma_{e_B}=p_Tp_n\,r_{T,T+1}r_{n,n+1}=1$. □

Example 2 For the vertex of Example 1, R_v has fifteen vertices and twelve edges, and consists of three paths and one cycle. The extreme creases are e_3, at $70°$, and e_2, at $-30°$, and the sides are (f_3) and $(f_4, f_5, f_6, f_1, f_2)$. The path of the extreme creases consists of the four rows in which the largest sector f_3 is the tortilla, (e_1, f_3), (e_6, f_3), (e_5, f_3), (e_4, f_3); it runs from $\{2,3\}$ through $\{1,3\}, \{3,6\}, \{3,5\}$ to $\{3,4\}$, every vertex with one sector on each side, and forces e_2 and e_3 equal. The sectors f_1 and f_5 $(40°)$ are smaller than both their neighbours, and the rows $(e_6, f_2), (e_1, f_6)$ and $(e_4, f_6), (e_5, f_4)$ are paths of length two on the second side, pairing the minimum e_6 with the maximum e_1 and the minimum e_4 with the maximum e_5, both forced opposite. The remaining rows (e_1, f_4), (e_4, f_2), (e_1, f_5), (e_4, f_1) form a cycle on the pairs $\{1,4\}, \{2,4\}, \{2,5\}, \{1,5\}$, also on the second side. The state of Fig. 2, MMMMVV, satisfies the three relations, with mountains at the extreme creases and $M - V = 2$. The sixteen solutions of the rows induce the $2^3 = 8$ assignments that satisfy them, all with $M - V = \pm 2$, and these are also the assignments of the twelve flat-folded states (Corollary 1). Four of the sixteen solutions are not transitive, and satisfy Maekawa's rule all the same.

Proposition 1 (Nested pairs) *At an interior vertex without taco–taco rows, the pairs of Theorem 2, drawn as chords between the creases on a circle, do not cross: the creases that lie strictly between the two creases of a pair, on either side of it, are paired among themselves.*

The proof, by crimping, is in Appendix B. In Example 2 the pairs $\{e_4, e_5\}$ and $\{e_6, e_1\}$ lie side by side on the second side; Figure 3(c) shows pairs nested three deep.

Corollary 1 (Flat-foldable assignments) *At a generic interior vertex of degree $2m$, a mountain–valley assignment is flat-foldable if and only if it is induced by an assignment of the layer variables that satisfies the taco–tortilla rows at the vertex, if and only if it satisfies the m relations of Theorem 2. There are 2^m such assignments.*

Proof $W_{\text{flat}}(v) \subseteq W_{\text{lin}}(v)$, and, by Lemma 3 and Theorem 2, $W_{\text{lin}}(v)$ is the set of the 2^m assignments that satisfy the relations. Hull's recursion gives a generic vertex exactly 2^m flat-foldable assignments, a factor two for every crimp [13]: in his terms no two consecutive angles are equal at any step of the crimping, since crimping removes crease positions without moving the others. The sets therefore coincide. □

Proposition 2 (Rank at a generic vertex) *At a generic interior vertex with N overlapping pairs of sectors, let c be the number of cycles of R_v. The taco–tortilla rows have rank $N-m-c$, hence 2^{m+c} solutions, and each of the 2^m flat-foldable assignments is induced by exactly 2^c of them.*

Proof Every row involves two variables, so the rows are the edges of R_v, and since they have solutions their rank is the number of vertices of R_v minus the number of its components, $N - m - c$. A solution is fixed by one bit on every component; the bit of a path fixes the assignments of its two creases, and the bit of a cycle fixes no assignment. □

For the vertex of Example 1, $N = 15$, $m = 3$ and $c = 1$, and the rank is 11, as found there. Corollary 1 and Proposition 2 also hold at every vertex without taco–taco rows, with the same proofs: there no two consecutive angles are equal, at any step of Hull's recursion. Corollary 1 is a special case of Corollary 2 below, and Appendix B gives a second proof of Theorem 2, by crimping.

Persistence.

Which creases are paired can be read off the walk alone. Regard the angle of the folded image as a height on the circle of the paper around the vertex, and sweep a level t upwards. The points of height at most t form arcs of the circle: an arc is born at every minimum e_k when $t = c_k$, and when t reaches a maximum the arcs on its two sides merge or, at the highest crease, the last arc closes up into the circle. By the *elder rule* of persistent homology [16, 17], when two arcs merge at a maximum e_M, the one whose lowest point is higher dies, and its minimum e_m forms with e_M a *persistence pair*, of persistence $c_M - c_m$. Every minimum other than the lowest crease is paired in this way, and the lowest and the highest crease form the *essential* pair [18]. At a vertex without taco–taco rows no two minima, and no two maxima, lie on one ray, so the rule meets no tie; a minimum and a maximum on one ray are not adjacent, and do not interact.

Theorem 3 (Persistence) *At an interior vertex without taco–taco rows, the pairs of Theorem 2 are the persistence pairs of the angle of the folded image, read as a height function on the circle of the paper around the vertex; the pair of the extreme creases is the essential pair.*

The proof, again by crimping, is in Appendix B: a sector of smallest angle is a pair of smallest persistence, whose arc dies at once, and crimping it changes no other pair. Hull's recursion [13], which crimps a smallest sector at every step, therefore cancels the persistence pairs in order of increasing persistence, the angle of each crimped sector being the persistence of its pair. The theorem can be read as a filling of lakes (Fig. 3c): water poured at a minimum fills its basin until it overflows, at the maximum of its pair, into the basin of a lower minimum. A minimum inside the basin overflows earlier, at a maximum inside it, which is Proposition 1 again. In Example 2 the sweep meets the minima e_2, e_6 and e_4 at $-30°$, $0°$ and $10°$; at $40°$ the maximum e_1 merges the arcs of e_6 and e_2, and e_6 dies; at $50°$ the maximum e_5 merges the arc of e_4 into that of e_2; and at $70°$ the circle closes at e_3. The persistence pairs are $\{e_6, e_1\}$, $\{e_4, e_5\}$ and the essential pair $\{e_2, e_3\}$, the pairs of Example 2.

7 All vertices

At a degenerate vertex the taco–tortilla and taco–taco rows no longer decide which assignments fold flat (Examples 3 and 4). They decide exactly the affine hull of the flat-foldable assignments.

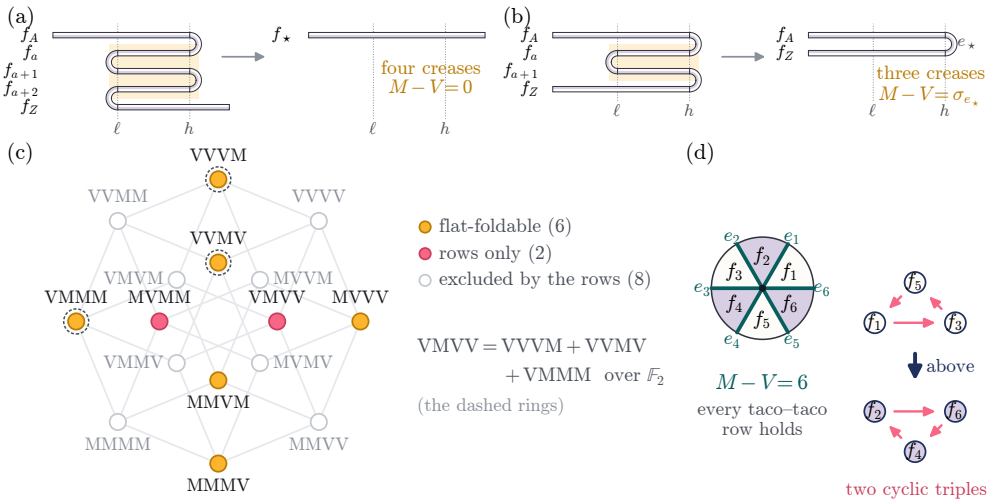


Fig. 4 Degenerate vertices. (a), (b) Crimping a run of equal smallest sectors, seen edge-on over the angular coordinate as in Fig. 2(b); the run and its interval $J = [\ell, h]$ are shaded. (a) A run of three sectors ($k = 2$): f_A reaches J from one side and f_Z leaves it on the other, and crimping merges f_A, the run and f_Z into one sector $f_\star$. The run's four creases have $M - V = 0$ in every flat-folded state. (b) A run of two sectors ($k = 1$): f_A and f_Z both lie on the same side of h, and crimping replaces the run's three creases by one crease $e_\star$ at h. In every flat-folded state the three creases have $M - V = \sigma_{e_\star}$. (c) The sixteen assignments $\sigma_{e_1}\sigma_{e_2}\sigma_{e_3}\sigma_{e_4}$ of the vertex of Example 3 on the 4-cube, an edge changing one crease. The rows allow the eight with $M - V = \pm 2$, the affine hull over $\mathbb{F}_2$ of the six flat-foldable ones (saffron); the two others (raspberry) are sums of three flat-foldable ones, for instance VMVV = VVVM + VVMV + VMMM. (d) The regular vertex of Example 4: its six taco–taco rows allow every crease to be a mountain, with every odd sector above every even one and each triple ordered cyclically, which no layer order does

Crimping runs.

Let w be a closed walk whose angles are not all equal, α its smallest angle, and $f_a, \ldots, f_{a+k}$, with $k \geq 0$, a maximal run of consecutive sectors of angle α, so that $f_A = f_{a-1}$ and $f_Z = f_{a+k+1}$ have larger angles. The sectors of the run share one interval $J = [\ell, h]$, and their $k+2$ creases $e_{a-1}, \ldots, e_{a+k}$ lie alternately at the two ends of J. *Crimping* the run (Fig. 4) gives a closed walk w'. If k is even, f_A, the run and f_Z merge into one sector $f_\star$ of angle $\alpha_A - \alpha + \alpha_Z$, and the $k+2$ creases disappear. If k is odd, the run disappears, f_A and f_Z keep their angles, and the $k+2$ creases are replaced by one crease $e_\star$ between them. All other sectors and creases keep their angles and positions. Crimping repeatedly, every walk reduces to one whose angles are all equal; let $E(w)$ be the number of crimps with k even. For $k = 0$ a single sector is crimped, as in Appendix B, and crimps, with end folds, are the basic moves of one-dimensional flat folding [19].

Hull's recursion.

Hull's analysis of crimping describes the flat-foldable assignments of a vertex, of a sheet or of a cone, recursively [13, Thms. 3–5]. The $k+2$ creases of a crimped run have $M - V = 0$ when k is even and $M - V = \pm 1$ when k is odd, and then the run folds as

pair of sectors of w	k even	k odd
both in Q	β	$[\{f_A, f_Z\}]$
f_A or f_Z with an outer f_g	$[\{f_\star, f_g\}]$	$[\{f_A, f_g\}]$ or $[\{f_Z, f_g\}]$
a run sector with an outer f_g reaching h	$[\{f_\star, f_g\}]$	$[\{f_A, f_g\}]$
a run sector with an outer f_g not reaching h	$[\{f_\star, f_g\}]$	undefined
two outer sectors	$[\{f_g, f_{g'}\}]$	$[\{f_g, f_{g'}\}]$

Table 1 The map ψ in the proof of Lemma 4. Q is the set formed by f_A, the run and f_Z; the other sectors are outer; an outer sector reaches h if h is interior to its interval or is its upper endpoint; $[u]$ is the component of $H_{w'}$ containing the vertex u

the single crease $e_\star$, with $\sigma_{e_\star} = M - V$; the other creases, together with $e_\star$ when k is odd, carry a flat-foldable assignment of w'; every combination of such choices is flat-foldable; and at a vertex with all angles equal the flat-foldable assignments are those with $M - V = \pm 2$. We write $W_{\mathrm{Hull}}(v)$ for the assignments produced by this recursion, so that $W_{\mathrm{Hull}}(v) = W_{\mathrm{flat}}(v)$; the proof of Theorem 4 uses the inclusion $\subseteq$. We write $W_4(v)$ for the assignments produced by the same recursion with every condition taken modulo four: $M - V \equiv 0$ or $M - V \equiv \sigma_{e_\star} \pmod 4$ on the creases of each crimped run, and $M - V \equiv 2 \pmod 4$ at the end.

Theorem 4 *At every interior vertex v of degree n,*

$$W_{\mathrm{lin}}(v) = \mathrm{aff}_{\mathbb{F}_2} W_{\mathrm{flat}}(v) = W_4(v), \qquad \dim W_{\mathrm{lin}}(v) = n - 1 - E(v) :$$

the mountain–valley assignments induced by the solutions of the taco–tortilla and taco–taco rows at v are the affine hull over $\mathbb{F}_2$ of the flat-foldable ones, and they are given by Hull's recursion read modulo four. Equivalently, the hypergraph of the rows has $\gamma(v) = E(v) + 1$ components containing adjacent pairs.

Since $n - 1 - E(v)$ is the dimension of $W_{\mathrm{lin}}(v)$, the number $E(v)$ does not depend on which run of smallest sectors is crimped at each step. The theorem rests on a lemma about crimping, in which every closed walk carries the rows of Lemma 1 and the hypergraph of Sec. 5.

Lemma 4 (Crimping a run) *Let w' be obtained from w by crimping a maximal run of sectors of smallest angle. Then $\gamma(w) \geq \gamma(w') + 1$ when k is even and $\gamma(w) \geq \gamma(w')$ when k is odd.*

Proof Write Q for the set of sectors f_A, $f_a, \ldots, f_{a+k}$, f_Z, and call the others *outer*. Replacing every c_k by $-c_k$ if needed, we may assume that e_{a-1} lies at h, a maximum. Since α is the smallest angle and the run is maximal, $I_A = [c_{a-2}, h]$ with $c_{a-2} < \ell$; if k is even, e_{a+k} lies at ℓ and $I_Z = [\ell, c_{a+k+1}]$ with $c_{a+k+1} > h$, so that $I_\star = [c_{a-2}, c_{a+k+1}]$ contains J in its interior; if k is odd, e_{a+k} lies at h, $I_Z = [c_{a+k+1}, h]$ with $c_{a+k+1} < \ell$, and $e_\star$ lies at h. An outer interval is at least as long as J; if it has an endpoint inside J, it reaches beyond the other end of J. The creases of the run are the only creases of w at ℓ or h incident to Q, since c_{a-2} and c_{a+k+1} lie outside $[\ell, h]$.

We define a map ψ from vertices of H_w to components of $H_{w'}$, or to an extra symbol β when k is even, with three properties: (1) the vertices of every row of w at which ψ is defined have the same image; (2) a row containing a vertex at which ψ is undefined contains no vertex at which it is defined; (3) every adjacent pair of w' is the image of an adjacent pair of w, and, when k is even, the adjacent pairs of the run's creases are sent to β. By (1) and (2), every component of H_w is sent to a single component of $H_{w'}$ or to β, and by (3) distinct components of $H_{w'}$ containing adjacent pairs, and β, come from distinct components of H_w containing adjacent pairs. This gives the inequalities. Table 1 summarises ψ.

k even. Let $\psi(\{x, y\}) = \beta$ if $x, y \in Q$; $\psi(\{x, g\}) = [\{\star, g\}]$ if $x \in Q$ and g is outer; and $\psi(\{g, g'\}) = [\{g, g'\}]$ for outer g, g'. These are vertices of $H_{w'}$, since $I_\star$ contains the intervals of Q. Property (2) is empty and (3) is clear, the creases e_{a-2} and e_{a+k+1} becoming the creases of $f_\star$. For (1) we go through the rows of w. A row involving no sector of Q is a row of w', since the outer sectors and their creases are unchanged. A taco–tortilla row (e_j, f_t) with an outer taco and $t \in Q$ has $c_j \in \operatorname{int} I_t \subseteq \operatorname{int} I_\star$, so $(e_j, f_\star)$ is a row of w' on the two images. A row (e_j, f_g) with g outer and e_j a crease of the run sends both its vertices to $[\{\star, g\}]$; with $e_j = e_{a-2}$ or e_{a+k+1} it is a row of w', the taco becoming $(f_{a-2}, f_\star)$ or $(f_\star, f_{a+k+2})$. Rows at the run's creases with tortilla in Q lie in β, and there are no rows at e_{a-2} or e_{a+k+1} with tortilla in Q, these creases lying outside the intervals of Q other than their own sectors. A taco–taco row between two outer creases, or between e_{a-2} or e_{a+k+1} and an outer crease, is a row of w'; one between two creases of the run lies in β; one between a crease of the run and an outer crease e_x at ℓ or h has the images $[\{\star, x\}]$ and $[\{\star, x+1\}]$, which the row $(e_x, f_\star)$ of w' joins, since $c_x \in \operatorname{int} I_\star$; and none joins a crease of the run to e_{a-2} or e_{a+k+1}, which lie outside $[\ell, h]$.

k odd. If an outer sector g reaches h, then $\{f_A, f_g\}$ and $\{f_Z, f_g\}$ lie in one component of $H_{w'}$: when h is interior to I_g the row $(e_\star, f_g)$ joins them, and when h is the upper endpoint of I_g the upper crease of f_g, a maximum at h, and $e_\star$ have a common taco–taco row containing both. Let $\psi(\{x, y\}) = [\{A, Z\}]$, the component of the adjacent pair of $e_\star$, if $x, y \in Q$; $\psi(\{b, g\}) = [\{A, g\}]$ if b is in the run and g is outer and reaches h, and undefined if g does not reach h; and $\psi(u) = [u]$ for the remaining pairs, which are pairs of sectors of w' with the same intervals. Property (3) is clear. Rows involving no sector of the run and no crease of the run are rows of w'. For the others:

- a row at a crease of the run with tortilla in Q lies in $[\{A, Z\}]$;
- a row (e_c, f_g) at a crease of the run with g outer: if e_c lies at h, then g reaches h and both vertices go to $[\{A, g\}] = [\{Z, g\}]$; if e_c lies at ℓ, both sectors of the taco are in the run, and both vertices go to $[\{A, g\}]$ or are both undefined;
- a row (e_j, f_b) with b in the run and an outer taco $f_g, f_{g'}$ has c_j inside J; if c_j is a minimum, both outer sectors extend upwards beyond h, and their images $[\{A, g\}]$, $[\{A, g'\}]$ are joined by the row (e_j, f_A) of w', since $c_j \in \operatorname{int} I_A$; if c_j is a maximum, neither reaches h, and both vertices are undefined;
- a taco–taco row between a crease of the run and an outer crease e_x: at h the outer sectors extend downwards from h, so they reach it, and all images lie in the component of the taco–taco row $(e_x, e_\star)$ of w', which contains $\{f_A, f_x\}, \{f_Z, f_x\}, \{f_A, f_{x+1}\}, \{f_Z, f_{x+1}\}$; at ℓ the crease of the run has both sectors in the run, the outer sectors extend upwards from ℓ and reach h, and the images $[\{A, x\}]$, $[\{A, x+1\}]$ are joined by the row (e_x, f_A) of w';
- a taco–taco row between two creases of the run lies in $[\{A, Z\}]$, and none joins a crease of the run to e_{a-2} or e_{a+k+1}, which lie below ℓ.

A vertex $\{b, g\}$ with g not reaching h lies only in rows of the second and third kinds, whose vertices are then all undefined; this is property (2). □

Proof of Theorem 4 By Lemma 3, $\dim W_{\text{lin}}(v) = n - \gamma(v)$. Crimping v down to a walk with all angles equal, where $\gamma \geq 1$, Lemma 4 gives $\gamma(v) \geq E(v)+1$, so $\dim W_{\text{lin}}(v) \leq n-1-E(v)$. On the other hand $W_{\text{Hull}}(v) \subseteq W_{\text{flat}}(v) \subseteq W_{\text{lin}}(v)$, and $W_{\text{lin}}(v)$ is affine, so $\operatorname{aff} W_{\text{Hull}}(v) \subseteq \operatorname{aff} W_{\text{flat}}(v) \subseteq W_{\text{lin}}(v)$. It remains to see that $\operatorname{aff} W_{\text{Hull}}(v) = W_4(v)$ and that this space has dimension $n-1-E(v)$; then all the inclusions are equalities.

Both follow by induction along the crimps from one fact: for $L \geq 2$ creases and $0 < V_0 < L$, the assignments with exactly V_0 valleys span, over $\mathbb{F}_2$, the affine space of the assignments with $V \equiv V_0 \pmod 2$, of dimension $L-1$; and $V \equiv V_0 \pmod 2$ is $M - V \equiv L - 2V_0 \pmod 4$. If k is even, the assignments of v given by the recursion are the pairs formed by an assignment of the run's $k+2$ creases with $M - V = 0$ and one of v', so their affine hull is the product of the parity class of the run, of dimension $k+1$, with $\operatorname{aff} W_{\text{Hull}}(v')$; the run adds $k+2$ creases and $k+1$ dimensions. If k is odd, the run's assignments allowed by each value of $\sigma_{e_\star}$ span the corresponding parity class, and the map that sends an assignment of v to its restriction together with the value of $\sigma_{e_\star}$ fixed by the run's parity is affine and onto, with fibres that are translates of one subspace of dimension $k+1$; the affine hull is the preimage of $\operatorname{aff} W_{\text{Hull}}(v')$, and the run adds $k+1$ creases and $k+1$ dimensions. At the end, the assignments of d equal angles with $M - V = \pm 2$ span those with $M - V \equiv 2 \pmod 4$, of dimension $d-1$. Adding up, $\operatorname{aff} W_{\text{Hull}}(v) = W_4(v)$ has dimension $n-1-E(v)$. □

Blocks.

Call *blocks* the sets of creases that the recursion treats together: the $k+2$ creases of a run crimped with k even form a block; when k is odd they take the place of $e_\star$ in its block of w'; and the creases of the final walk, with all angles equal, form one block. There are $E(v)+1$ blocks, and $W_4(v)$ prescribes the parity of the number of valleys in each of them. By properties (1) and (2) in the proof of Lemma 4, every component of H_w is sent to a single component of $H_{w'}$ or to β, so two creases whose adjacent pairs lie in one component of H_v lie in one block. The $\gamma(v)$ components containing adjacent pairs therefore refine the $E(v)+1$ blocks, and by Theorem 4 they are as many: they are the blocks. In particular the blocks, and not only their number, do not depend on which runs are crimped.

Remark 1 (A proof without Hull's theorem) The proof above uses Hull's analysis through the inclusion $W_{\text{Hull}}(v) \subseteq W_{\text{flat}}(v)$. Two elementary facts about crimps can replace it, together with the computation of $\operatorname{aff} W_{\text{Hull}}(v)$. Write $\delta_k \in \mathbb{F}_2^n$ for the assignment that is 1 at e_k and 0 elsewhere. First, if $n \geq 4$ and f_a is no longer than its two neighbours, a flat-folded state of the walk obtained by crimping f_a unfolds to one of v, with f_{a-1}, f_a, f_{a+1} stacked in zigzag in the place of $f_\star$, in either orientation of the stack; the two states of v differ exactly at e_{a-1} and e_a, so $\delta_{a-1} + \delta_a$ is a direction of $\operatorname{aff} W_{\text{flat}}(v)$. Second, the crimp of a run is a sequence of such crimps of single sectors, and unfolding them with fixed orientations sends the flat-folded states of w' to flat-folded states of w through an affine map of the assignments, which keeps the values on the creases of w', with $e_\star$ standing for e_{a-1} when k is odd, and is constant on the other creases; it carries directions of $\operatorname{aff} W_{\text{flat}}(w')$ to directions of $\operatorname{aff} W_{\text{flat}}(w)$. By induction along the recursion, $\delta_i + \delta_j$ is a direction of $\operatorname{aff} W_{\text{flat}}(v)$ whenever e_i and e_j lie in one block: at the end, every sector of a walk of degree at least four with all angles equal can be crimped, and

at degree two the two flat-folded states differ at both creases. Now let $\sigma_0 \in W_{\text{flat}}(v) \cap W_4(v)$, for instance the assignment obtained by unfolding along the recursion. Lemma 3, with the refinement of the blocks by the components, gives $W_{\text{lin}}(v) \subseteq W_4(v)$, since the parity of the number of valleys in a block is the sum of the quantities that the equations of its components fix. And $W_4(v)$, which is σ_0 plus the assignments with an even number of ones in every block, lies in $\operatorname{aff} W_{\text{flat}}(v)$ by the directions above. Hence $W_4(v) \subseteq \operatorname{aff} W_{\text{flat}}(v) \subseteq W_{\text{lin}}(v) \subseteq W_4(v)$, and the $E(v)+1$ blocks give $\dim W_4(v) = n-1-E(v)$. Hull's enumeration [13] is then needed only for the converse in Corollary 2.

Comparing with Hull's count of flat-foldable assignments [13] gives the exact place where transitivity matters.

Corollary 2 (Where the linear constraints suffice) $|W_{\text{lin}}(v)| = 2^{n-1-E(v)}$, *and* $W_{\text{lin}}(v) = W_{\text{flat}}(v)$ *if and only if every run crimped in the recursion is a single sector and the recursion ends at degree two or four. In particular* $W_{\text{lin}}(v) = W_{\text{flat}}(v)$ *at every vertex without taco–taco rows.*

Proof Along the recursion, Hull's count multiplies by $\binom{k+2}{\lfloor (k+2)/2 \rfloor}$ at a crimp with k even and by $\binom{k+2}{(k+1)/2}$ at one with k odd, and ends with $2\binom{d}{d/2-1}$ at d equal angles [13]; Theorem 4 multiplies by 2^{k+1} and ends with 2^{d-1}. The two agree for $k=0$ and for $d=2,4$, and the binomials are smaller otherwise. At a vertex without taco–taco rows no two crease positions of the same type coincide, at any step of the recursion, so every crimped run is a single sector and the recursion ends at degree two. □

Example 3 At the degree-four vertex with angles $45°, 135°, 135°, 45°$ (Fig. 4c), the creases e_1 and e_3 lie on one ray, and the rows are (e_4, f_2), (e_4, f_3) and the taco–taco row (e_1, e_3); they form a single component, so $\gamma = 1$ and W_{lin} is the set of the 8 assignments with $M - V = \pm 2$. The two sectors of $45°$ form a run with $k=1$, and Hull's recursion requires $M - V = \sigma_{e_\star} = \sigma_{e_2}$ on the creases e_3, e_4, e_1: 6 assignments fold flat. The other two, VMVV and MVMM, satisfy Maekawa's rule and every linear constraint, but would fold the two small sectors with three creases of the same type.

Example 4 (The regular vertex) At the vertex of degree six with all angles $60°$ (Fig. 4d), the odd creases lie on one ray of the folded image and the even ones on another, and every sector has the same interval. There is no taco–tortilla row, and there are six taco–taco rows, one for each pair of odd creases and each pair of even ones. Here $E = 0$, and W_{lin} is the set of the $2^5 = 32$ assignments with $M - V \equiv 2 \pmod 4$, of which 30, all but the two with every crease of the same type, are flat-foldable. The all-mountain assignment comes from an explicit solution: put every odd sector above every even one, and order each triple cyclically,

$$f_1 > f_3 > f_5 > f_1, \qquad f_2 > f_6 > f_4 > f_2,$$

where $f > f'$ means that f lies above f'. The six rows hold: for the pair (e_1, e_3), for instance, $r_{13} r_{23} r_{14} r_{24} = (+1)(-1)(+1)(-1) = 1$. By Eq. (1) every crease is a mountain. Theorem 1 holds here with no taco–tortilla row at all, as in its proof, e_6 being at the lowest ray: written in the orientation $i < j$, the six rows contain every adjacent pair once, every other pair twice,

and four reversed variables, $x_{21}, x_{31}, x_{41}, x_{51}$, two in each of the rows (e_2, e_6) and (e_4, e_6), so their sum is $\sum_{k<6} x_{k,k+1} + x_{16} = 0$, which is the vertex row $\sum_k x_{k,k+1} = 1$ once x_{16} is written as $x_{61} + 1$.

8 Consequences

Admissible perturbations.

If some crease rays coincide, move the positions c_k slightly, keeping every maximum a maximum and every minimum a minimum; the walk still closes, so Kawasaki's condition holds, and a common rescaling restores the total angle. Choose the perturbation so that, whenever a maximum and a minimum coincided, the minimum moves to the right of the maximum: their sectors, which extended in opposite directions, still do not overlap, and no row is created between them. Every taco–tortilla row of the original vertex is a row of the perturbed one, and the same pairs of sectors overlap in both. The only new rows come from coinciding creases e_k, e_l of the same type: if both are maxima and c_l moves to the left of c_k, so that e_l moves into the sectors of e_k, these sectors f_k and f_{k+1} now cover c_l, and the two new rows are

$$x_{lk} + x_{l+1,k} = 0, \qquad x_{l,k+1} + x_{l+1,k+1} = 0. \tag{6}$$

The four sectors are distinct, e_k and e_l being of the same type, and the sum of the two rows is the taco–taco row of (e_k, e_l), since reversing the four pairs adds $4 \equiv 0$; minima are symmetric, with e_l moving to the right. By Lemma 1 every taco–taco row at v arises this way, exactly once. We call such a perturbation *admissible*; on the creases of one ray and one type, it is determined, as far as rows are concerned, by the order in which they move into one another's sectors.

Maekawa's theorem.

A flat-folded state of a degenerate vertex satisfies more than the linear rows: transitivity orders the folds that share a ray. This is all it takes to reach a generic vertex.

Proposition 3 (Flat-folded states and perturbations) *Every flat-folded state of an interior vertex v satisfies the taco–tortilla rows of an admissible perturbation of v.*

Proof Let a flat-folded state of v be given, and call the size of the taco of a crease e_k the number of layers between f_k and f_{k+1} (Fig. 5a). Take an admissible perturbation in which, among the creases of one ray and one type, those with smaller tacos move into the sectors of those with larger ones, ties being broken arbitrarily. Its rows are the rows of v, except that every taco–taco row (e_k, e_l) is replaced by the two rows (6), with e_l the crease that moves into the sectors of e_k, whose taco is therefore not smaller. These say that f_k and f_{k+1} each lie on one side of the pair f_l, f_{l+1}, and they hold. By the taco–taco row, f_k and f_{k+1} lie both between f_l and f_{l+1} or both outside them, so that the two tacos are nested or stacked (Fig. 1b); and both between would make the taco of e_k smaller than that of e_l. □

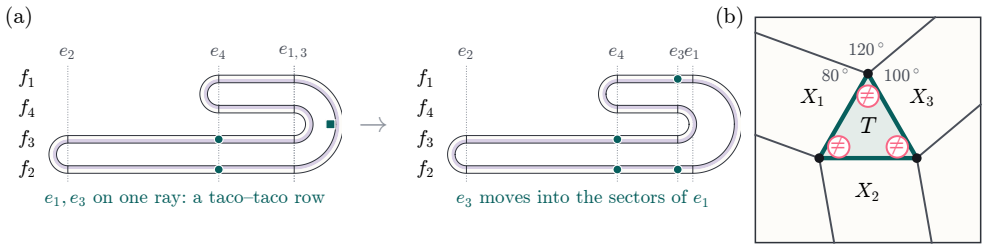


Fig. 5 Consequences. (a) Proposition 3 at the vertex of Example 3: a flat-folded state, f_1, f_4, f_3, f_2 from top to bottom, in which the maxima e_1 and e_3 lie on one ray and have a taco–taco row (square); and the admissible perturbation in which e_3, whose taco is the smaller, moves into the sectors of e_1. The same order satisfies the taco–tortilla rows of the perturbed vertex (dots). (b) The triangle pattern of Sec. 8: three degree-four vertices with angles $60°, 100°, 120°, 80°$, whose smallest sectors form the triangle T. By the big–little–big lemma the two edges of T at each corner have opposite assignments, which no assignment of the three edges satisfies; the six rows of the three lemmas add up to $0 = 1$

Corollary 3 (Maekawa's theorem from the taco constraints) *Let v be an interior vertex.*

(a) Every assignment of the layer variables between the sectors at v that satisfies the taco–tortilla and taco–taco rows at v has $M - V \equiv 2 \pmod 4$, and $M - V = \pm 2$ if v has degree four or no taco–taco row.
(b) Every flat-folded state of v has $M - V = \pm 2$.

Proof (a) follows from Theorem 1 with Eq. (3), and from Theorem 2. For (b), Proposition 3 makes the state a solution of the taco–tortilla rows of a generic vertex with the same creases, and Theorem 2 applies. □

None of these proofs uses Maekawa's theorem, and the only use of transitivity is the order of Proposition 3. Theorem 4 shows that the linear constraints alone give no more than (a): at the regular vertex of degree six they allow $M - V = 6$.

Big–little–big.

The shortest pair of Theorem 2 is the big–little–big lemma.

Proposition 4 (Big–little–big) *If the sector f_a is smaller than both its neighbours, the taco–tortilla rows (e_{a-1}, f_{a+1}) and (e_a, f_{a-1}) exist, and their sum is $r_{a-1,a} = r_{a,a+1}$; by Eq. (1), $\sigma_{e_{a-1}} \neq \sigma_{e_a}$.*

Proof I_a is shorter than I_{a+1}, with which it shares the endpoint c_a, so the other endpoint c_{a-1} lies inside I_{a+1}; symmetrically c_a lies inside I_{a-1}. The two rows read $r_{a-1,a+1} = r_{a,a+1}$ and $r_{a,a-1} = r_{a+1,a-1}$, whence $r_{a,a+1} = r_{a-1,a+1} = -r_{a+1,a-1} = -r_{a,a-1} = r_{a-1,a}$. With $p_{a-1} = -p_a$, Eq. (1) gives $\sigma_{e_{a-1}}\sigma_{e_a} = -r_{a-1,a}r_{a,a+1} = -1$. □

This is the classical proof of the lemma [2, 12], that otherwise the small sector would be covered on the same side by both its neighbours, written in layer variables: the two rows are the two coverings. It holds at every vertex, generic or not.

Whole crease patterns.

Every row at a vertex is a row of the whole pattern. The assignment induced by any solution of the linear system of a crease pattern therefore lies in $W_4(v)$ at every interior vertex v, by Theorem 4, and satisfies the pairs of Theorem 2 at every vertex without taco–taco rows.

Corollary 4 (No new information in the vertex rows) *Adding the vertex rows of all interior vertices to the $\mathbb{F}_2$-linear system of taco–tortilla and taco–taco rows of a pattern changes neither its rank nor its set of solutions. In particular, a parity certificate of non-foldability, a set of rows whose left-hand sides cancel and whose constants add up to one, never needs a vertex row.*

This follows from Theorem 1; by Theorem 4 and Lemma 3, the same holds for every relation that defines $W_4(v)$, such as the big–little–big relations, since each is a sum of rows at v. The linear system, which does not mention the mountain–valley assignment, already contains every such obstruction. At vertices without taco–taco rows the relations are the pairs of Theorem 2, and they form a signed graph on the creases of the pattern; a cycle of this graph whose signs multiply to -1 is a local obstruction to flat folding, and the rows of its paths add up to a certificate, as in the cycle certificates of Ref. [6]. Finding such a cycle takes linear time, as does deciding local flat foldability in general [8, 19]; the paths turn the obstruction into a certificate made of non-penetration constraints. The classical pattern of three degree-four vertices around a triangle, with the smallest sector of each vertex inside it, is an example (Fig. 5b). It satisfies Kawasaki's condition everywhere and is locally flat-foldable at every vertex, but by the big–little–big lemma the three edges of the triangle would have to alternate between mountain and valley, which an odd cycle forbids [12]. Its linear system is inconsistent, and a certificate is given by six taco–tortilla rows, the two rows of Proposition 4 at each corner, whose left-hand sides cancel and whose constants add up to one.

9 Discussion

The mountain–valley assignment and the layer order carry the same information across a crease, and the local rules of flat folding, stated on the former, are consequences of the non-penetration constraints on the latter. Big–little–big is the sum of two linear constraints, the parity of Maekawa's rule the sum of all of them, and the full rule follows from the pairing of the creases once transitivity has ordered the folds that share a ray. At an arbitrary vertex the linear constraints see exactly the affine hull of the flat-foldable assignments over $\mathbb{F}_2$, which is Hull's crimping recursion with every count replaced by its parity. At a single vertex, transitivity, where the computational hardness of flat folding lies, does nothing else: it turns the parity of each crimped run back into its count, and it matters only where several sectors are stacked on one wedge of the folded image.

The proofs of Theorems 1 and 2 read the layer order from the extreme creases. In the Ising language of Ref. [5], numbering the sectors from a lowest crease is a gauge

in which every constraint at the vertex is ferromagnetic, and the holonomy $W_v = -1$ of the vertex row sits on the one adjacent pair written backwards. The reading also has a simple geometric meaning. Parametrize the paper around the vertex by a point s of a circle, and let $\gamma(s)$ be the angle of its folded image. The pairs of points with the same image, $\{s, s'\}$ with $\gamma(s) = \gamma(s')$, form a curve Z in the space of unordered pairs of points of the circle, a Möbius band whose boundary is the diagonal. An overlapping pair of sectors is a segment of Z; a taco–tortilla row is a point where one of the two points crosses a crease, a taco–taco row a point where Z crosses itself, and an adjacent pair an end of Z on the boundary, at a crease. The layer variable says which of the two points lies above, and the rows say that it does not change along Z. Lemma 2 says that Z is a curve, crossing itself at the taco–taco rows, with its ends at the creases; at a generic vertex it is R_v, and the admissible perturbations resolve its crossings. The extreme creases cut the band into two triangles, the pairs of points on one side, and a rectangle, the pairs with one point on each side. On each side of the rectangle one of the two points is at an extreme crease, so $\gamma(s) - \gamma(s')$ vanishes there only at the two corners on the diagonal, and Z joins them: this is the path of the extreme creases. The other paths are disjoint arcs in the triangles with their ends on the diagonal, so that on each side the creases they pair are nested like parentheses: this is Proposition 1 (Fig. 3). Which ends each arc joins is Theorem 3: the arc that leaves the diagonal at a minimum returns to it at the maximum where the sublevel arc of that minimum dies, or, for the lowest crease, at the highest.

Two other curves around the vertex measure the parity of Theorem 1 (Appendix A). The cross-section of the folded paper turns once and never crosses itself, and gives the classical proof; the arc diagram crosses itself and turns m times, and gives, through Whitney's formula, the parity of $M - V$ in every solution of the rows (Remark 2). The first needs a layer order; the second does not.

Four questions remain. For a whole crease pattern, the assignments induced by the solutions of the linear system satisfy the conditions of Theorem 4 at every vertex; are they exactly the affine hull of the locally flat-foldable assignments, and if not, what separates the two? At a degenerate vertex, which transitivity constraints, among the triples of sectors stacked on one wedge, suffice to cut the affine hull down to the flat-foldable assignments? At a vertex with taco–taco rows the components of H_v are the blocks of Hull's recursion, and the sweep of the heights meets ties; do the blocks have a description in terms of persistence? And our results concern patterns with convex facets, where the tortilla–tortilla constraints of Ref. [3] are absent; with non-convex facets those constraints appear, and the corresponding statements remain to be found.

Declarations

- **Funding.** No funding was received.
- **Competing interests.** The author has no competing interests to declare that are relevant to the content of this article.
- **Ethics approval and consent.** Not applicable.
- **Data availability.** No datasets were generated or analysed in this study.

- **Code availability.** The software `linfold`, which computes the rows, the relations and Hull's recursion at every vertex of a crease pattern, solves the linear system of the whole pattern and returns the parity certificates of Sec. 8, and a formalization of the results in the Lean 4 proof assistant [20], based on Mathlib [21], are available at https://github.com/andrea-ladiana/flat-folding-linear-constraints [22]. The formalization takes the rows of Lemma 1 as definitions, and does not cover the converse in Corollary 2. The scripts of Appendix C are available from the author.
- **Use of AI tools.** The proofs of this paper were transcribed into Lean with the assistance of an AI system (Claude, Anthropic).
- **Author contribution.** Not applicable (single author).

Appendix A Three other proofs of Theorem 1

The three proofs below write the layer variables in a fixed orientation and compute the constant of the sum of the taco–tortilla rows at a generic vertex. Degenerate vertices follow by an admissible perturbation (Sec. 8): every taco–tortilla row of v is a row of the perturbed vertex, every taco–taco row of v is the sum of the two rows that replace it, Eq. (6), and no other row is created, so the sum of the rows at v is the sum of the taco–tortilla rows of the perturbed vertex, which is its vertex row; and the vertex row is the same for both.

The linear part.

A taco–tortilla row (e_k, f_g) contains the pairs $\{k, g\}$ and $\{k+1, g\}$. The pair of sectors $\{a, b\}$ therefore occurs in the row (e_k, f_b) for every endpoint c_k of I_a lying inside I_b, and symmetrically, so its number of occurrences in the sum of all taco–tortilla rows is

$$\#\{\text{endpoints of } I_a \text{ inside } I_b\} + \#\{\text{endpoints of } I_b \text{ inside } I_a\}. \tag{7}$$

For a generic vertex this is 0 if the intervals are disjoint; 2 if they overlap and are not adjacent, whether they are nested $(2+0)$ or *staggered*, each containing one endpoint of the other $(1+1)$; and 1 for adjacent sectors (Fig. 6a). Indeed I_a and I_{a+1} share the endpoint c_a, which is interior to neither, and extend from it in the same direction, so exactly one other endpoint, that of the shorter one, lies inside the longer one. Summing over $\mathbb{F}_2$, the variables of non-adjacent pairs cancel, and the linear part of the sum is $\sum_k x_{k,k+1}$, that of the vertex row.

The constant.

To keep track of orientations we multiply instead of adding. The taco–tortilla row (e_k, f_g) is $r_{kg} r_{k+1,g} = 1$, and in the product of the left-hand sides an occurrence of the pair $\{a, b\}$ contributes the factor r_{ab} when an endpoint of I_a lies inside I_b, and $r_{ba} = -r_{ab}$ when an endpoint of I_b lies inside I_a. A nested pair contributes $r_{ab}^2 = 1$, a staggered pair $r_{ab} r_{ba} = -1$, and the adjacent pair $\{a, a+1\}$ contributes $r_{a,a+1}$ if I_a is

the shorter and $-r_{a,a+1}$ if I_{a+1} is. Hence, identically in the layer variables,

$$\prod_{(e_k,f_g)} r_{kg}\, r_{k+1,g} = (-1)^{S+D} \prod_{a} r_{a,a+1} = (-1)^{S+D}\, W_v, \tag{8}$$

where S is the number of staggered pairs of non-adjacent sectors and D the number of indices a with $|I_{a+1}| < |I_a|$. In additive form, the sum of the taco–tortilla rows at a generic vertex is the equation $\sum_a x_{a,a+1} = S + D$, which is the vertex row exactly when $S + D$ is odd; each of the three proofs below shows that it is.

Lemma 5 (Parity) *For every generic vertex, $S + D$ is odd.*

First proof: Whitney's formula Mark the crease positions $c_1, \ldots, c_n$ on a horizontal line and take $\delta > 0$ smaller than any distance between them. For every sector f_k draw the upper half-circle with diameter from $c_{k-1} + \delta$ to c_k, traversed in this direction, and at every crease e_k the lower half-circle of diameter δ from c_k to $c_k + \delta$. The pieces join with matching vertical tangents into a closed curve Γ with continuously turning tangent, the *arc diagram* of the folded vertex (Fig. 6b): the arc of f_k arrives at c_k going down, whether c_k is a maximum or a minimum, and the arc of f_{k+1} leaves $c_k + \delta$ going up.

Rotation. An odd sector is traversed from left to right, so its half-circle turns by $-\pi$; an even sector from right to left, turning by $+\pi$; and each of the n small half-circles below the line turns by $+\pi$. The rotation number of Γ is therefore $n\pi/2\pi = m$.

Double points. The small half-circles meet nothing. Two upper half-circles have their centres on the line and distinct endpoints, so they meet at most once above the line, transversally, and they do so exactly when their diameters interleave. For non-adjacent sectors this happens when their intervals are staggered, which gives S double points. For the adjacent sectors f_k and f_{k+1} the diameters are I_k and I_{k+1} with their common endpoint c_k split into c_k and $c_k + \delta$: at a maximum they interleave when I_{k+1} is the shorter, at a minimum when it is the longer. If A and B are the numbers of maxima and of minima at which the second sector is the shorter, so that $D = A + B$, the junctions contribute $J = A + (m - B) \equiv D + m$ double points modulo two. Since S, D and J depend only on the order of the c_k, a small move of the c_k removes triple points without changing them.

Whitney's formula [14] gives the rotation number of a closed curve with continuously turning tangent and transverse double points as a signed count of the double points plus ± 1; modulo two, the rotation number has the parity of the number of double points plus one. Hence $m \equiv S + D + m + 1$, and $S + D$ is odd. □

Second proof: wall crossing The angle vectors of closed walks (4) of length n form a convex cone, on which the walk depends linearly; the non-generic walks lie on the hyperplanes $c_k = c_l$, and on none of them is $l = k \pm 1$, since the angles are positive. Two generic walks are joined by a segment that crosses these hyperplanes one at a time, away from their intersections. We show that crossing one of them does not change the parity of $S + D$, and then compute it at one walk.

The status of a pair of sectors, disjoint, nested or staggered, changes only when an endpoint of one interval passes an endpoint of the other, and the comparison $|I_{a+1}| < |I_a|$ only when c_{a-1} passes c_{a+1}. At a crossing of c_k and c_l, therefore, only the four pairs with one sector in $\{k, k+1\}$ and the other in $\{l, l+1\}$ change status, and D changes only if $l = k \pm 2$.

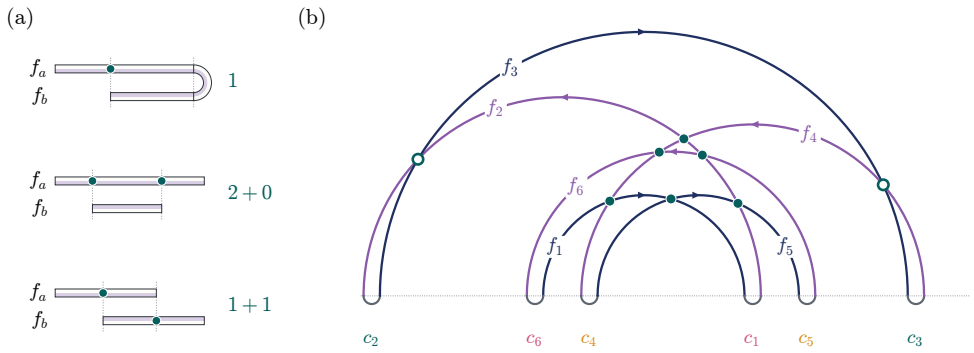


Fig. 6 The first proof of Lemma 5. (a) How often a pair of sectors occurs in the taco–tortilla rows, which by Eq. (7) is the number of endpoints of each that lie inside the other (dots): once if the sectors are adjacent, twice if they are nested or staggered. (b) The arc diagram Γ of the vertex of Fig. 2, on the same angular axis as Fig. 2(b): every sector an upper half-circle over its interval, traversed as the arrows show (the odd sectors from left to right, in navy, the even ones from right to left, in plum), and every crease a small half-turn below the axis. Its double points are the $S = 6$ staggered pairs of non-adjacent sectors (filled) and $J = 2$ junctions of adjacent ones (open), and its rotation number is $m = 3$; here $D = 3$. Whitney's formula, $3 \equiv 8 + 1 \pmod 2$, fixes the constant of Theorem 1. Unlike the cross-section of Fig. 2(b), Γ crosses itself and does not depend on the layer order

If one crease is a maximum and the other a minimum, the two pairs of sectors extend from the crossing point in opposite directions, and the four pairs pass from disjoint to staggered or back. None of them is adjacent, since $l - k$ is odd and $l \neq k \pm 1$, so S changes by four and D does not change.

If both creases are maxima, or both minima, all four sectors extend from the crossing point in the same direction, all four pairs overlap on both sides of the crossing, and each passes from nested to staggered or back. When no pair is adjacent, S changes by an even number and D does not change. When $l = k + 2$ and $n \geq 6$, exactly one pair, $\{k + 1, k + 2\}$, is adjacent: S changes by an odd number, and D by one, because the two intervals of that pair share the endpoint c_{k+1} and the crossing exchanges which is the longer. At degree four, $l = k + 2$ makes two pairs adjacent, and S and D both change by even numbers. In every case $S + D$ keeps its parity.

At the spiral

$$c_{2j} = j \quad (0 \leq j < m), \qquad c_{2j-1} = L + 1 - j \quad (1 \leq j \leq m), \qquad L > 2m,$$

the intervals $I_1 \supset I_2 \supset \cdots \supset I_{n-1}$ are nested and shrink by one at each step, $|I_k| = L+1-k$, while $I_n = [0, L + 1 - m]$ is staggered with $I_2, \ldots, I_{n-2}$ and adjacent to I_1 and I_{n-1}. Hence $S = n-3$ and $D = n-2$, the descents being $a = 1, \ldots, n-2$, and $S+D = 2n-5$ is odd. □

Third proof: one flat-folded state By Eq. (8), $S+D$ is the value of $\sum_a x_{a,a+1}$ at any solution of the taco–tortilla rows. Crimp a sector f_a of smallest angle, which merges f_{a-1}, f_a and f_{a+1} into one sector $f_\star$ (Appendix B), and unfold a flat-folded state of the crimped walk by putting f_{a-1}, f_a, f_{a+1}, stacked in zigzag, in the place of $f_\star$ in the layer order; this is a flat-folded state of v [13], in which $x_{a-1,a} = x_{a,a+1}$ and the other adjacent variables are those of the crimped walk. By induction on n, down to $n = 2$ where $x_{12} + x_{21} = 1$, the state has $\sum_a x_{a,a+1} = 1$, and $S + D$ is odd. □

Remark 2 (Two curves) The classical proof of Maekawa's rule follows the cross-section of the folded paper around a small circle centred at the vertex [11, 15]. Drawn with its layers slightly apart, as in Fig. 2(b), it is a simple closed curve, which turns by π at every crease, in one sense at mountains and in the other at valleys, and whose rotation number is ± 1; hence $M - V = \pm 2$. The arc diagram is a different curve. It does not depend on the layer order, it crosses itself, and its rotation number m is fixed by the walk; Whitney's formula then gives the parity of its crossings, and through Eq. (8) the parity of $M - V$ in every solution of the rows, whether or not it is a layer order.

Appendix B A proof of Theorem 2 by crimping

We give a second proof of Theorem 2 at generic vertices, by induction on the degree. The proof crimps a smallest sector. In the folded image, crimping f_a merges f_{a-1}, f_a and f_{a+1} into one sector; on the walk it removes c_{a-1} and c_a, and the result is a closed walk of length $n - 2$ whose angles add up to less than 2π, the vertex of a cone. We keep the names and positions of the other creases, and call $f_\star$ the merged sector, with angle $\alpha_{a-1} - \alpha_a + \alpha_{a+1}$ and side $p_\star = p_{a-1} = p_{a+1}$.

Lemma 6 (Crimping a sector) *Let v be generic of degree $n \geq 4$, f_a a sector of smallest angle, and v' the walk obtained by crimping f_a, which is generic.*

(a) The rows (e_{a-1}, f_{a+1}) and (e_a, f_{a-1}) form a path P_a of R_v, which pairs e_{a-1} with e_a and forces opposite assignments.
(b) If $n \geq 6$, the other $m - 1$ paths of R_v pair the same creases as the paths of $R_{v'}$, and force the same relations.

Proof Since v is generic, f_a is strictly smaller than both its neighbours. Replacing every c_k by $-c_k$ if needed, which changes no row, we may assume that c_{a-1} is a maximum. Then $c_{a-2} < c_a < c_{a-1} < c_{a+1}$, with $I_{a-1} = [c_{a-2}, c_{a-1}]$, $I_a = [c_a, c_{a-1}]$, $I_{a+1} = [c_a, c_{a+1}]$, and the merged sector of v' has $I_\star = [c_{a-2}, c_{a+1}] = I_{a-1} \cup I_{a+1}$.

(a) The endpoint c_{a-1} of I_a lies inside I_{a+1}, and c_a inside I_{a-1}, which gives the two rows. Both contain the pair $\{a-1, a+1\}$, which by Eq. (7) has no other row, and the adjacent pairs $\{a-1, a\}$ and $\{a, a+1\}$ have one row each. Adding the rows, $x_{a-1,a+1} + x_{a,a+1} + x_{a,a-1} + x_{a+1,a-1} = 0$ becomes $x_{a-1,a} + x_{a,a+1} = 0$, and $p_{a-1}p_a = -1$, so the relation is opposite.

(b) By Eq. (7), the rows that contain the pairs $\{a-1, g\}$, $\{a, g\}$ and $\{a+1, g\}$ of v, for a sector $g \notin \{a-1, a, a+1\}$, are of three kinds: rows at the creases e_{a-2} or e_{a+1} with tortilla f_g, which are rows of v' with the pair $\{\star, g\}$ in place of $\{a-1, g\}$ or $\{a+1, g\}$; rows at e_{a-1} or e_a with tortilla f_g, which disappear in v'; and rows at an end crease e_j of f_g with tortilla f_{a-1}, f_a or f_{a+1}, which become the single row $(e_j, f_\star)$ of v'. The interval I_g is at least as long as I_a and has no endpoint in common with it, so it is not contained in I_a, and one of four cases holds.

(i) I_g does not meet the interior of I_a. Then $\{a, g\}$ does not overlap, and I_g overlaps at most one of I_{a-1} and I_{a+1}, which lie on either side of I_a outside it. The rows of that pair are rows of v' with $f_\star$ in its place.

(ii) I_a lies in the interior of I_g. The rows (e_{a-1}, f_g) and (e_a, f_g) join $\{a-1, g\}$, $\{a, g\}$ and $\{a+1, g\}$ in a chain, and say $r_{a-1,g} = r_{a,g} = r_{a+1,g}$; the middle pair has no other row.

Contracting the chain to the single pair $\{\star, g\}$ of v' leaves the remaining rows of $\{a-1, g\}$ and $\{a+1, g\}$, which are those of $\{\star, g\}$.

(iii) I_g contains c_{a-1} but not c_a in its interior, so that its lower endpoint is a crease position c_j inside I_a. This is a minimum, at which f_g meets a sector f_h whose interval also starts at c_j; being at least as long as I_a, it reaches beyond c_{a-1}, so f_h is of the same type, and $h \notin \{a-1, a, a+1\}$, since c_j is not an endpoint of their intervals. The rows (e_{a-1}, f_g), (e_j, f_a), (e_{a-1}, f_h) and (e_j, f_{a-1}) join the pairs $\{a-1, g\}$, $\{a, g\}$, $\{a, h\}$, $\{a-1, h\}$ in a cycle, and by Eq. (7) these four pairs have no other row: the cycle is a component of R_v, and it disappears in v'. The third row at c_j, (e_j, f_{a+1}), becomes $(e_j, f_\star)$, and $\{a+1, g\}$, $\{a+1, h\}$ become $\{\star, g\}$, $\{\star, h\}$.

(iv) I_g contains c_a but not c_{a-1} in its interior, so that its upper endpoint is a crease position c_j inside I_a. This is a maximum, at which f_g meets a sector f_h of the same type, and the rows (e_a, f_g), (e_j, f_a), (e_a, f_h) and (e_j, f_{a+1}) form a cycle on the pairs $\{a+1, g\}$, $\{a, g\}$, $\{a, h\}$, $\{a+1, h\}$, which is a component of R_v and disappears in v'; the third row at c_j, (e_j, f_{a-1}), becomes $(e_j, f_\star)$, and $\{a-1, g\}$, $\{a-1, h\}$ become $\{\star, g\}$, $\{\star, h\}$. The argument is that of (iii), reflected.

When $g = a-2$ the same holds, with the adjacent pair $\{a-2, a-1\}$ of v corresponding to the adjacent pair $\{a-2, \star\}$ of v'; here I_{a-2} starts at $c_{a-2} < c_a$, so case (iii) does not occur, and in case (iv) $h = a-3$, which is not $a+1$ because $n \geq 6$. The case $g = a+2$ is symmetric. Every row of v at e_{a-1}, at e_a or with tortilla f_a appears above: those with tortilla f_a are at crease positions inside I_a, which are endpoints of sectors of type (iii) or (iv).

Thus $R_{v'}$ is obtained from R_v by deleting the path of (a) and the cycles of (iii) and (iv), contracting the chains of (ii), and renaming $\{a \pm 1, g\}$ as $\{\star, g\}$. The adjacent pairs of v' are those of v other than $\{a-1, a\}$ and $\{a, a+1\}$, at the same creases, so every other path of R_v becomes a path of $R_{v'}$ with the same ends, and every path of $R_{v'}$ arises in this way. Each contraction identifies variables that the deleted rows set equal, so adding the rows of a path gives the same κ_P in v and in v', and since $p_\star = p_{a\pm1}$ the relation (5) is the same. □

Proof of Theorem 2 at generic vertices By induction on n, for generic closed walks. Let f_a be a sector of smallest angle. If $n = 4$, Lemma 6(a) gives one path, forced opposite, and the other path P' pairs the two remaining creases. No row joins two adjacent pairs when $n = 4$, so each path contains one of the two non-adjacent pairs, and R_v has no cycle. The sum of all the rows is then the sum of the rows of the two paths, $\sum_k x_{k,k+1} = \kappa_{P_a} + \kappa_{P'}$, and by Theorem 1 it is the vertex row, so $\kappa_{P_a} + \kappa_{P'} = 1$. Since $p_1 p_2 p_3 p_4 = 1$, Eq. (5) gives $\varepsilon_{P_a} \varepsilon_{P'} = -1$, and P' is forced equal. If $n \geq 6$, Lemma 6 shows that v has one more opposite pair than the generic walk v' of length $n-2$, and the same equal pairs. □

Crimping a smallest sector never removes an extreme crease: in the proof of Lemma 6, after the reflection made there, $c_{a-1} < c_{a+1}$ and $c_{a-2} < c_a$. The creases left at degree four are therefore the extreme creases of v, and the pair forced equal is formed by them, as in Theorem 2.

Proof of Proposition 1 At a vertex without taco–taco rows an admissible perturbation changes no row (Sec. 8), nor the cyclic order of the creases, and leads to a generic vertex; we may therefore assume v generic, and argue by induction on the degree. At degree two there is one pair. Otherwise let f_a be a sector of smallest angle. By Lemma 6, e_{a-1} and e_a form a pair; they are consecutive, so their chord crosses no other. The other pairs are those of the crimped walk, which is generic and keeps the other creases in the same cyclic order, so by induction they do not cross. □

Proof of Theorem 3 An admissible perturbation of a vertex without taco–taco rows changes no row (Sec. 8), and no persistence pair either: it only moves a minimum above a maximum that shared its ray, and the two, which are not adjacent, do not interact in the sweep. We may therefore assume v generic, and argue by induction on the degree. At degree two the two creases are the extreme ones, and form both the pair and the essential pair. Otherwise let f_a be a sector of smallest angle. By Lemma 6, e_{a-1} and e_a form a pair, and the other pairs are those of the crimped walk v', whose extreme creases are those of v.

The sector f_a is smaller than both its neighbours, so c_{a-1} and c_a lie strictly between c_{a-2} and c_{a+1}. Of e_{a-1} and e_a, call e_M the maximum and e_m the minimum, and let e_y be the other neighbour of e_M and e_z the other neighbour of e_m; then $c_y < c_m < c_M < c_z$. In the sweep, the arc born at e_m stays inside the two sectors at e_m until $t = c_M$, when it reaches e_M and merges with the arc that contains the points near e_y, of height below c_m. That arc has an older minimum, so $\{e_m, e_M\}$ is a persistence pair. Crimping replaces the three sectors between e_y and e_z by a single sector, along which the height runs monotonically from c_y to c_z, and keeps all other creases and heights. At every level, the arcs of v and of v' therefore contain the same creases of v'; the only other arc is that of e_m in v, for $c_m \le t < c_M$, which contains no crease of v'. The other minima are born and merge at the same levels in v and in v', with the same comparisons of age, so the persistence pairs of v other than $\{e_m, e_M\}$ are those of v', which by induction are its pairs. □

Appendix C Computations

Every statement of the paper was checked by computer, with Python scripts that are available from the author. Vertices with coinciding crease rays were generated with all angles integer multiples of one step, chosen among 10°, 12°, 15°, 18°, 20°, 30°, 36°, 45° and 60°, so that coincidences are exact.

Theorem 1 was checked with an implementation that folds the polygons of a pattern and extracts the rows from the folded images, independently of the angular picture used in the proof. For every interior vertex it sums the rows at that vertex and compares the sum with the vertex row. The identity holds on all 535 vertices tested: 240 single vertices of degree 4 to 14 with random angles; 240 single vertices of degree 4 to 12 whose angles are multiples of 15° or 30°, 198 of them with coinciding crease rays; and 55 vertices of named patterns, namely the interior vertices of map-folding grids, Miura-ori and pleated grids with up to 4×4 facets, the waterbomb and preliminary bases, the triangle pattern of Sec. 8, and three single vertices with repeated angles or of degree six. The six-row certificate of the triangle pattern was checked in the same way.

Lemma 5 was checked on 20000 random generic walks of degree 4 to 40. On 400 of degree 4 to 12 the arc diagram was built as a polygonal curve, and its rotation number and double points, computed from the drawing, were found equal to m and $S + J$.

Theorem 2 and Lemma 6 were checked on 8700 random generic walks of degree 4 to 40: in every case the row graph consists of cycles and m paths, exactly one path forces equal assignments, the path of Lemma 6(a) is a component forcing opposite ones, and the paths of the crimped walk pair the same creases with the same relations. The proofs of Theorems 1 and 2 were checked on 4001 random generic walks of degree 4 to 40 and on 6002 walks with integer angles, of degree 4 to 14, 5297 of them with taco–taco rows: in the numbering from a lowest crease every row is homogeneous, and

only the adjacent pair of that crease is written backwards. At the generic walks the rows never mix the three classes of pairs, the path of the extreme creases is the only one with a sector on each side and the only one forced equal, every path has an even number of rows and joins a maximum to a minimum, and on each side the paired creases are nested. *Theorem 3* was checked, by computing the persistence pairs with a union–find sweep, on 5000 random generic walks of degree 4 to 40 and on 3000 walks with integer angles and no taco–taco row, of degree 4 to 14; the order-only form of the elder rule used in the Lean formalization on 6000 further walks, 785 of them with coinciding crease rays; and the order in which Hull's recursion cancels the pairs on 6000 random generic walks of degree 4 to 48. *Corollary 1* was checked on 132 random generic vertices of degree 4, 6 and 8, by enumerating all layer orders.

Theorem 4 was checked on 13123 distinct vertices of degree 4 to 12, 11218 of them with taco–taco rows: in every case $W_{\mathrm{lin}}(v)$, computed by elimination, equals $W_4(v)$ and the affine hull of $W_{\mathrm{Hull}}(v)$. On 2313 of them, of degree at most six, and on 120 vertices of degree eight with taco–taco rows, the enumeration of all layer orders gave $W_{\mathrm{flat}}(v) = W_{\mathrm{Hull}}(v)$. *Lemmas 2, 3 and 4* were checked on 1901 vertices, together with $\gamma = E + 1$; the case analysis in the proof of Lemma 4 was checked row by row on 2677 crimps. *Corollary 2* was checked on 3321 vertices, and *Proposition 2* on the 423 generic ones among them. *Proposition 3* was checked on 327 vertices with taco–taco rows, 60 of them of degree eight, by enumerating all layer orders and all admissible perturbations.

Examples 1–4 were checked by enumerating all assignments of the layer variables.

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